

Quantitative Finance Project Portfolio

Machine Learning, Portfolio Risk and Alpha Research

Riccardo Cornelli

MSc Economics & Finance | Applied portfolio analytics and quantitative research

Purpose of this document

This single PDF consolidates three complete project reports into one professional portfolio for company applications, interviews and business-facing discussions. All original pages are preserved; the cover only provides structure and positioning.

Included reports

1 **Machine Learning Portfolio Construction & Risk Management**

Stock selection, ranking, Markowitz allocation, Monte Carlo and stress testing

7 pages

2 **Macro-Aware Portfolio Risk Analysis**

FactSet portfolio analytics, VaR/CVaR, macro regressions and constrained allocation

8 pages

3 **Bitcoin Intraday Market Microstructure Research**

1-minute BTC data, rare-event ML signals, walk-forward validation and trading rules

7 pages

Core capabilities demonstrated

Python / R data pipelines

Machine learning validation

Portfolio construction

Risk diagnostics

Macro-factor regression

Monte Carlo / stress testing

Machine Learning Portfolio Construction & Risk Management

Professional portfolio case study based on a master thesis project

Riccardo Cornelli

MSc Economics & Finance | Quantitative asset management project

What the project demonstrates

This portfolio summarizes a complete investment research pipeline: data engineering, supervised learning, walk-forward validation, stock ranking, portfolio construction, transaction cost analysis, Markowitz allocation, Monte Carlo simulation and stress testing. The objective is not to show isolated charts, but to make the full decision chain understandable from an asset-management and risk-management perspective.

S&P 500

investment
universe

17

technical features

Top 20% **2014-2025**

monthly stock
selection

out-of-sample
backtest

Core investment logic

Machine learning is used as a ranking engine, not as a perfect price forecaster. Forecasts are converted into cross-sectional scores; only the highest-ranked stocks are selected. The allocation layer then evaluates whether the selected universe can generate a superior risk-return profile versus passive benchmarks.

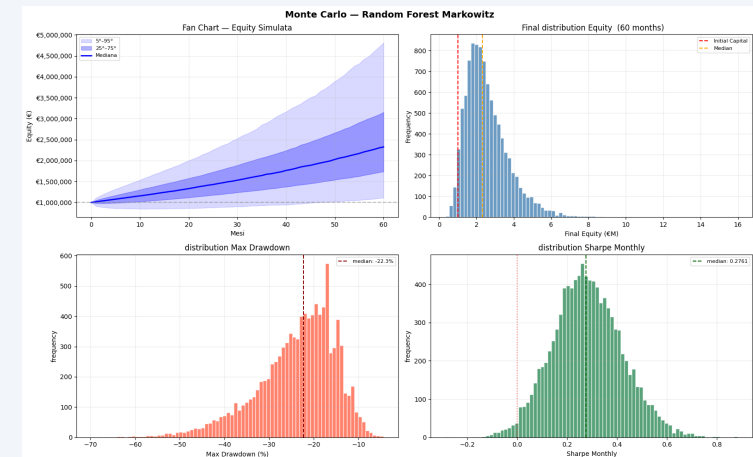


Exhibit: Monte Carlo distribution and path risk

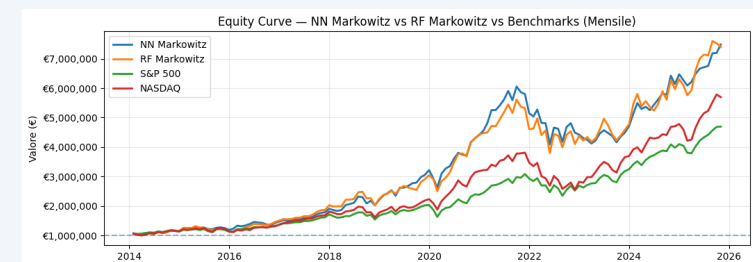


Exhibit: optimized portfolios and benchmarks

The empirical pipeline starts from the S&P 500 universe and builds a clean master long panel for model training and portfolio simulation.

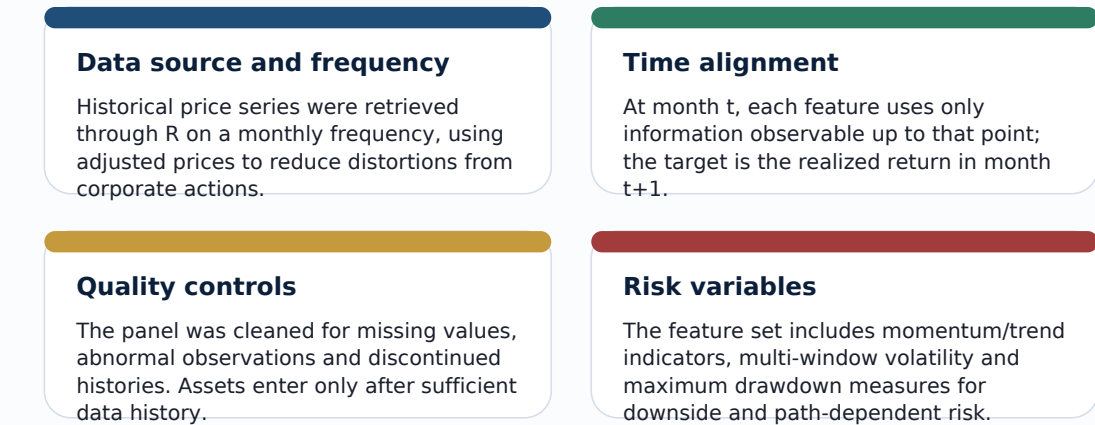
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Exhibit: sample of stock-level monthly features

- Using a ticker-month panel makes the process repeatable and auditable: every prediction can be traced back to the information available at that date.
- Monthly frequency reduces noise versus daily data and matches the investment horizon used for portfolio rebalancing.
- Drawdown and volatility features prevent the model from ranking stocks only by past growth; they add a risk-sensitive dimension to stock selection.

2. Machine learning layer: prediction as an asset-ranking signal

Random Forest and Neural Network models are embedded in the same monthly walk-forward protocol to avoid look-ahead bias and keep results comparable.

Random Forest

Tree-based ensemble used to capture nonlinear relationships and interactions between trend, volatility and drawdown variables. Selected because it delivered the strongest cross-sectional IC profile.

Neural Network

MLP regression architecture with imputation, standardization, ReLU hidden layers and early stopping. Used as a flexible nonlinear scoring model for complex return patterns.

Model comparison: error metrics vs ranking metrics

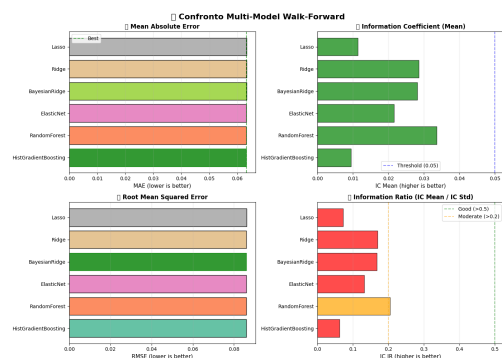


Exhibit: multi-model walk-forward comparison

Feature importance: which signals drive the ranking

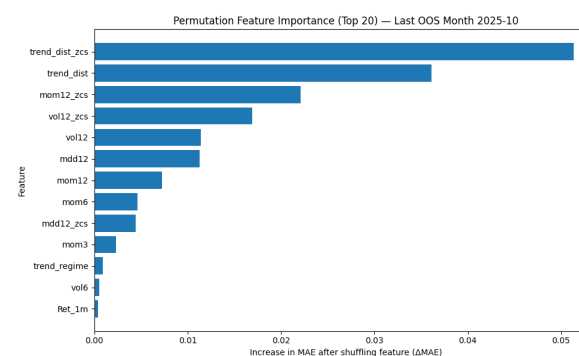


Exhibit: permutation importance on the final OOS month

9,967

OOS forecasts per tested model

0.0337

RF mean Information Coefficient

0.204

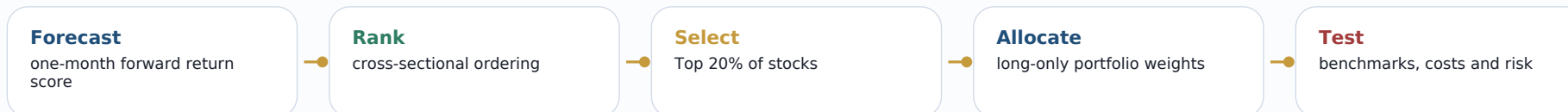
RF IC Information Ratio

98k+

NN out-of-sample predictions

3. From forecasts to investable portfolios

The models do not mechanically allocate capital to all stocks. They generate monthly scores, rank the cross-section, and select the highest-ranked names.



Operating rule

- At each month, all investable stocks receive a predictive score from the model.
- The portfolio keeps only the top 20% of ranked names; the rest of the universe is excluded.
- The portfolio is long-only and rebalanced monthly; transaction costs are modeled at 0.15% one-way.
- Benchmarks include the S&P 500 and NASDAQ to separate true stock-selection alpha from broad market exposure.

Predictive interpretation

The models are weak point forecasters, which is normal in equity-return prediction. Their value comes from ranking power: even a modest, persistent information signal can become economically useful once translated into a disciplined portfolio rule.

Prediction vs realized returns: weak point forecasts, useful ranking logic



Exhibit: prediction-realization scatter for NN and RF

0.0368

NN
prediction-realization
correlation

0.0298

RF
prediction-realization
correlation

56.2%

RF upper-tail hit
rate

108/213

NN positive-alpha
months

4. Backtest evidence: growth, drawdown and benchmark discipline

The strategies are evaluated against passive benchmarks to understand whether the ML ranking adds value beyond simple market exposure.

Performance and risk comparison across strategies

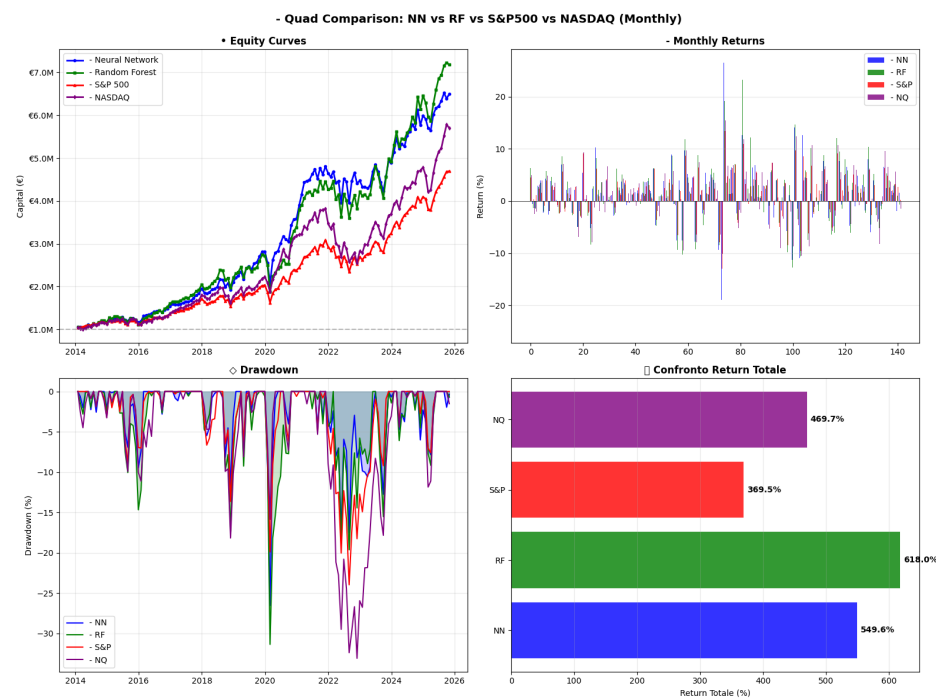


Exhibit: equity curve, returns, drawdown and performance bars

Backtest summary

Common sample: Jan 2014 - Oct 2025, monthly rebalancing

Strategy	Total return	Volatility	Max DD	Sharpe
NN Top 20%	549.57%	18.37%	-26.53%	0.93
RF Top 20%	618.04%	20.59%	-31.36%	0.88
S&P 500	369.51%	14.82%	-23.97%	0.94
NASDAQ	469%	n.a.	-33.1%	n.a.

Risk-return reading

Random Forest delivers the highest terminal wealth, but with higher volatility and deeper drawdowns. Neural Network gives up some return, but the trajectory is cleaner and closer to a professional risk-adjusted profile.

€7.18m

RF final capital

€6.50m

NN final capital

€4.70m

S&P 500 final capital

5. Optimization layer: machine-learning selection + Markowitz allocation

After selecting the Top 20% universe, the project adds a maximum-Sharpe mean-variance allocation to test whether risk-based weighting improves portfolio efficiency.

Optimized portfolios versus benchmarks

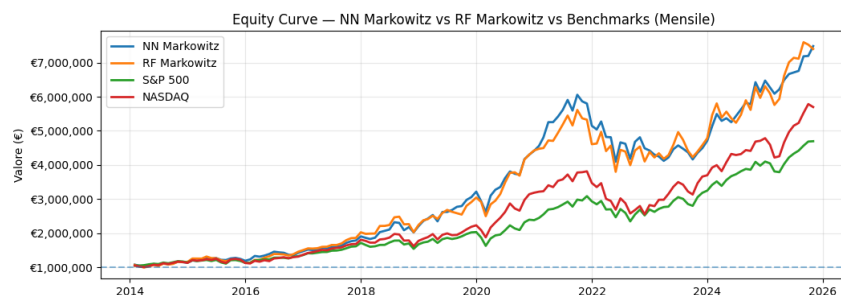


Exhibit: capital evolution of Markowitz strategies

Risk profile after allocation

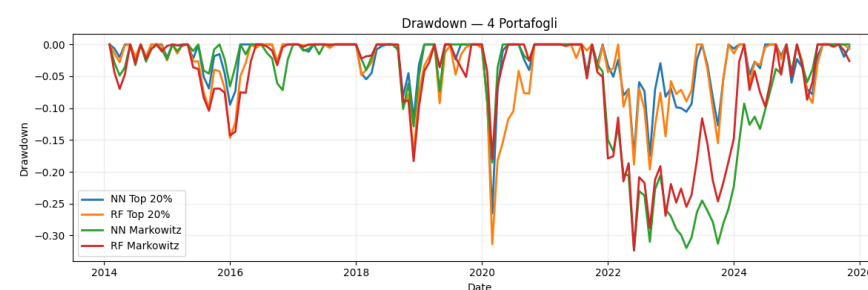


Exhibit: drawdown comparison across portfolios

Numerical impact of the optimization layer

Monthly metrics for the Top 20% strategies before and after Max-Sharpe allocation

Portfolio	Avg monthly return	Monthly vol.	Monthly Sharpe	Max DD	Total return	Turnover
NN Top 20%	1.464%	5.302%	0.2761	-26.53%	549.57%	n.a.
NN Markowitz	1.558%	5.140%	0.3031	-32.39%	648.34%	77%
RF Top 20%	1.574%	5.944%	0.2649	-31.36%	618.04%	n.a.
RF Markowitz	1.586%	5.799%	0.2736	-32.32%	639.96%	50%

Key insight

Optimization improves monthly Sharpe and reduces volatility, but it does not remove drawdown risk. The framework therefore also tracks turnover, transaction costs and scenario losses.

6. Robustness: path risk, stress testing and investment conclusions

The final layer evaluates whether the strategies remain credible beyond the historical equity curve by using simulation and crisis-period diagnostics.

Monte Carlo: NN Markowitz

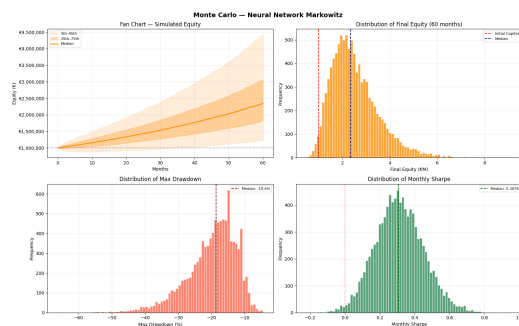


Exhibit: distribution of 10,000 paths over 60 months

Stress test: 2007-2009

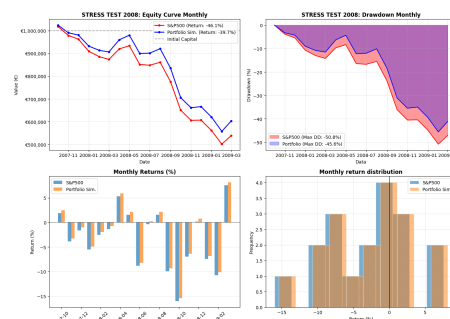


Exhibit: crisis-period portfolio behavior

Stress test: 2022

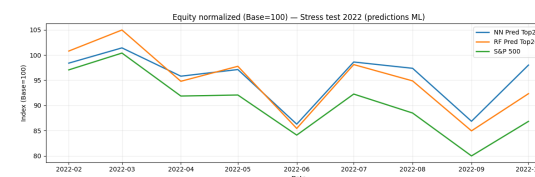


Exhibit: real-prediction stress window

10,000

bootstrap simulations

60m

forward simulation horizon

1.6%

NN probability below €1m

-7.67%

NN loss in 2022 stress test

-13.16%

S&P 500 loss in 2022 stress test

Portfolio value proposition

- End-to-end quant workflow: data preparation in R, ML modeling in Python, walk-forward validation, dynamic ranking and rebalanced portfolio construction.
- Professional risk reading: returns are interpreted together with volatility, Sharpe ratio, maximum drawdown, turnover, transaction costs, Monte Carlo dispersion and stress-test losses.
- Practical conclusion: the Neural Network appears more balanced in risk-adjusted terms, while Random Forest is more growth-oriented. Markowitz allocation improves efficiency but does not remove downside risk.

Selected thesis project | Ready for CV, LinkedIn and interview discussion

Macro-Aware Portfolio Risk Analysis

Explained project portfolio - Python, FactSet data and macro-factor regression

Project objective

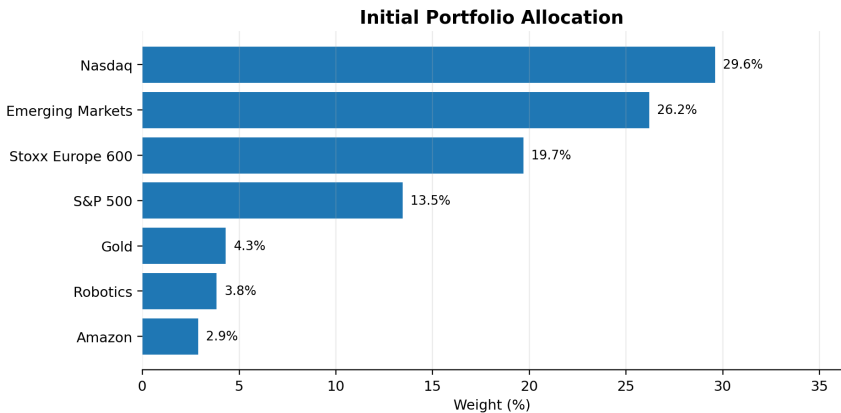
This project analyzes a real personal multi-asset portfolio built before a more advanced finance toolkit was developed. The goal was not simply to report performance, but to understand whether the allocation was genuinely diversified, which risks dominated the portfolio, how macroeconomic shocks affected returns, and how a constrained Markowitz + macro allocation framework could improve the decision process.

What the analysis actually did

- 1 Built the weighted portfolio return series from FactSet asset returns and actual invested amounts.
- 2 Measured historical performance, drawdowns, monthly VaR and CVaR at portfolio and asset level.
- 3 Created a monthly macro dataset with rates, inflation, VIX, yield curve and dollar variables.
- 4 Ran OLS regressions with HAC robust errors to estimate macro sensitivity and standardized beta exposure.
- 5 Implemented a constrained grid-search allocation combining Markowitz metrics and macro-risk penalties.

Input data and return construction

The starting point was the real portfolio allocation. Each asset return series was imported from FactSet files, cleaned, converted to numeric values, aligned on common dates and converted into decimal returns when necessary. The portfolio return was then computed as a weighted sum of the asset returns using the actual investment weights.



Core formula

Portfolio return: $R_{p,t} = \sum_i w_i \times R_{i,t}$. Monthly returns were obtained by compounding daily returns: $R_m = \text{Product}(1 + R_d) - 1$. This makes the analysis consistent across daily, monthly and yearly frequencies.

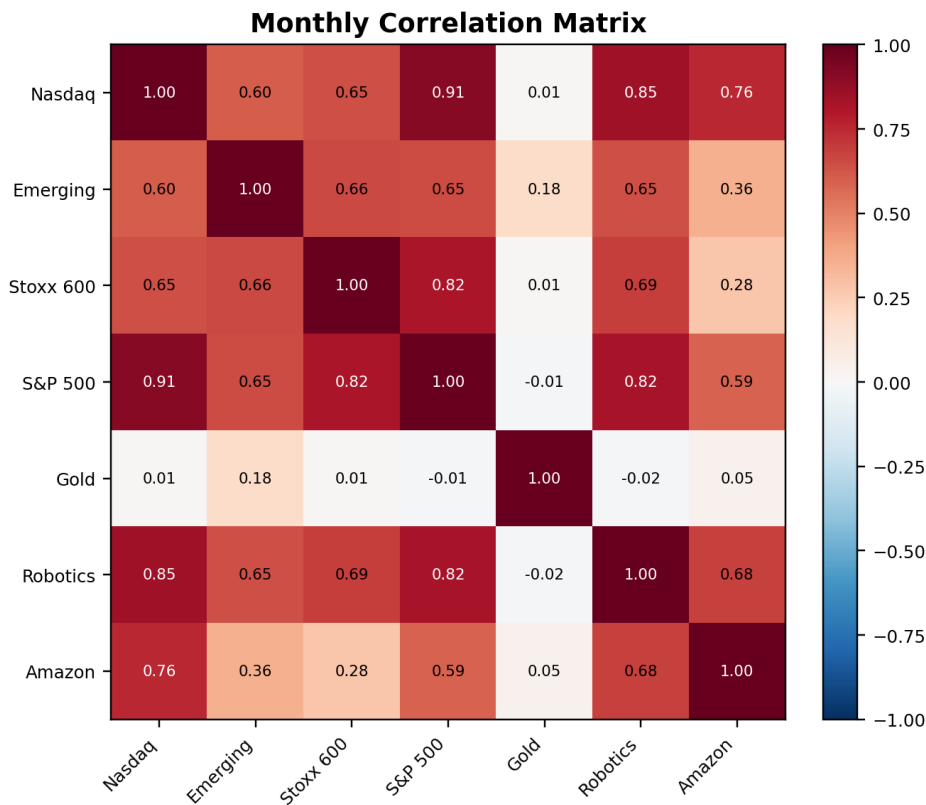
Risk and performance modules

Module	What it measures	Why it matters
Performance	Capital curve, return, Sharpe, Sortino	Return quality, not only final profit
Downside risk	Drawdown, VaR, CVaR	Loss severity and tail events
Diversification	Correlation matrix	Hidden common risk factors
Macro exposure	OLS regression and beta	Sensitivity to rates, VIX, inflation
Allocation	Markowitz + macro score	Improvement scenario under constraints

Asset	Initial weight
Nasdaq	29.62%
Emerging Markets	26.20%
Stoxx Europe 600	19.70%
S&P 500	13.46%
Gold	4.31%
Robotics	3.84%
Amazon	2.88%

Initial diagnosis

The allocation looked diversified by number of instruments, but the analytical question was whether it was economically diversified after accounting for correlations, downside risk and macro exposure.



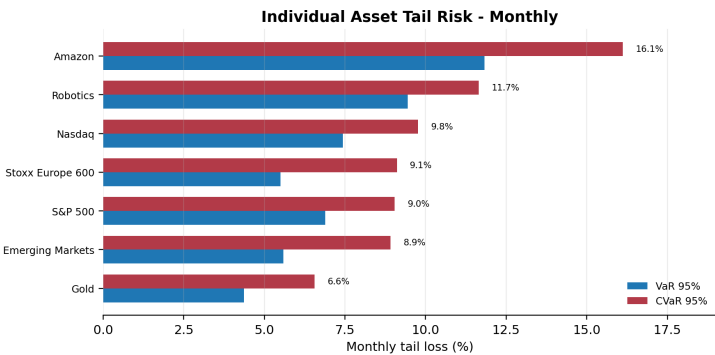
What the correlation matrix shows

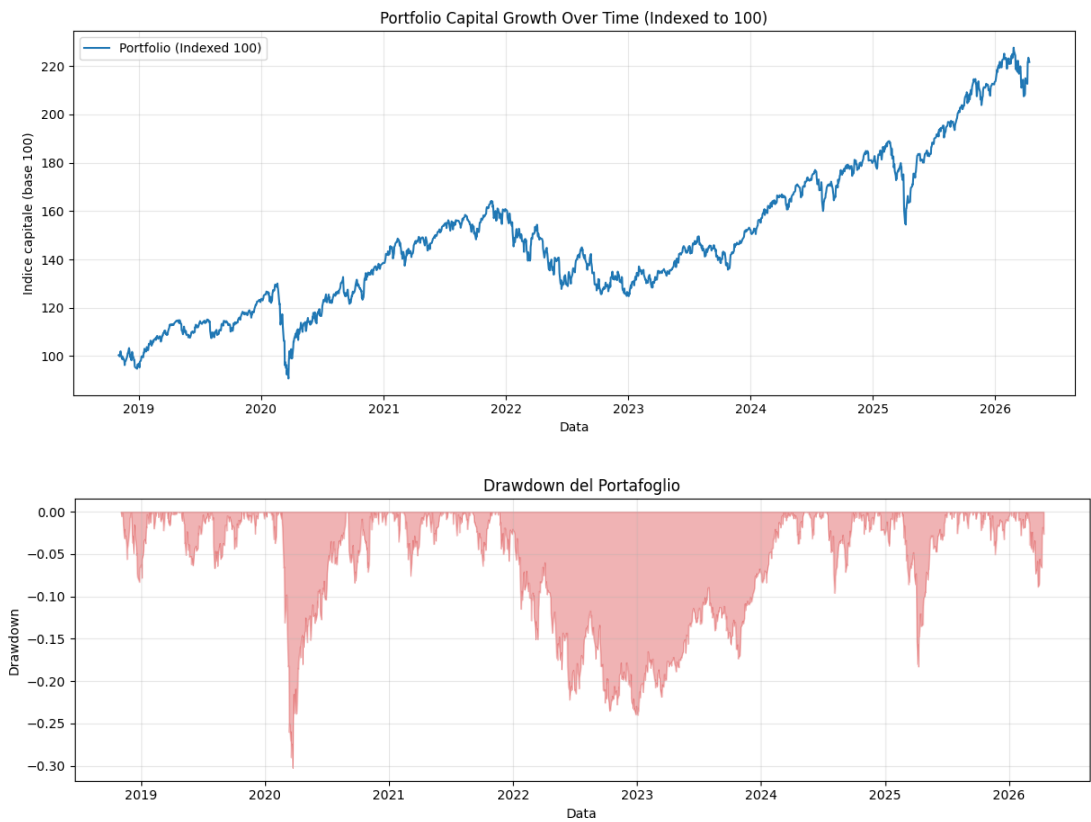
The portfolio was not equally diversified across independent risk sources. Nasdaq, S&P; 500, Robotics and Amazon show strong positive correlations, which means that several positions were likely to fall together during equity/growth sell-offs. Gold was the main low-correlation exposure: it had near-zero monthly correlation with Nasdaq and S&P; 500.

Relationship	Monthly corr.	Meaning
Nasdaq vs S&P 500	0.91	Very similar equity beta
Nasdaq vs Robotics	0.85	Growth/theme risk overlap
S&P 500 vs Robotics	0.82	High shared market factor
Gold vs S&P 500	-0.01	Potential diversifier
Gold vs Nasdaq	0.01	Low common movement

Which assets carried the largest tail risk?

To identify which positions were most fragile in downside months, individual monthly VaR and CVaR were computed. This complements the macro regression: the regression explains portfolio sensitivity to macro factors, while VaR/CVaR shows which assets had the largest standalone tail losses.





How VaR and CVaR were used

Historical VaR at 95% estimates the loss threshold exceeded in the worst 5% of observations. CVaR at 95% goes further: it measures the average loss inside that worst 5% tail. Therefore, CVaR is more useful when the objective is to understand how severe losses can be when the VaR threshold is breached.

Risk metric	Daily	Monthly	Yearly
VaR 95%	1.60%	6.79%	14.69%
CVaR 95%	2.57%	7.68%	22.11%
VaR 99%	3.19%	7.80%	20.62%
CVaR 99%	4.17%	9.00%	22.11%

Historical growth

>2.2x

indexed base 100

Worst drawdown

~ -30%

COVID shock

Monthly VaR 95%

6.79%

tail threshold

Monthly CVaR 95%

7.68%

avg. tail loss

Monthly obs.

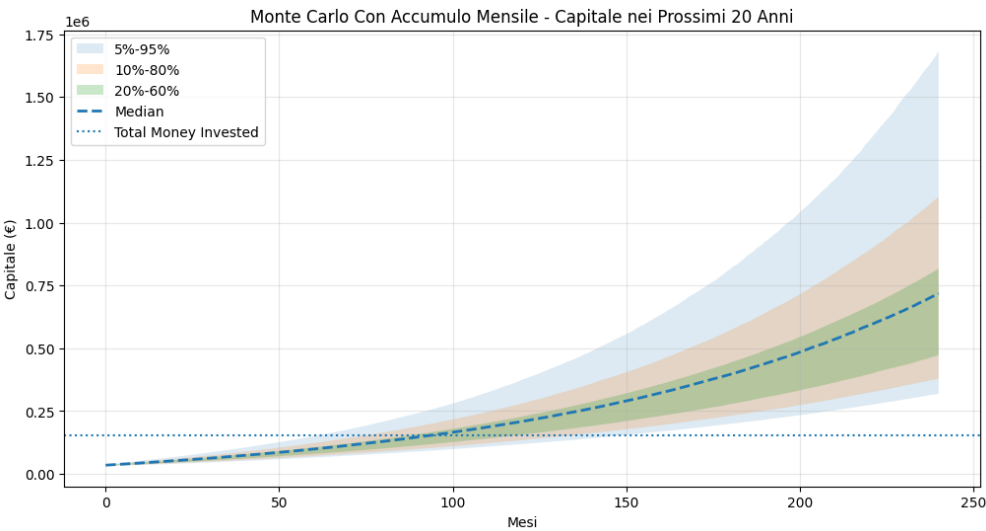
90

2018-2026 sample

Model setup

The Monte Carlo engine used a bootstrap approach: each simulated month randomly samples one historical monthly portfolio return, with replacement. The 20-year simulation starts from EUR 34,776 and adds EUR 500 at the end of each month after applying the sampled return. This produces a distribution of possible capital paths rather than a single deterministic forecast.

Input / output	Value
Number of simulations	50,000
Investment horizon	20 years / 240 months
Initial capital	EUR 34,776
Monthly contribution	EUR 500
Total money invested	EUR 154,776
Median final capital	EUR 718,144
Mean final capital	EUR 822,580
5th percentile	EUR 317,709
95th percentile	EUR 1,682,774



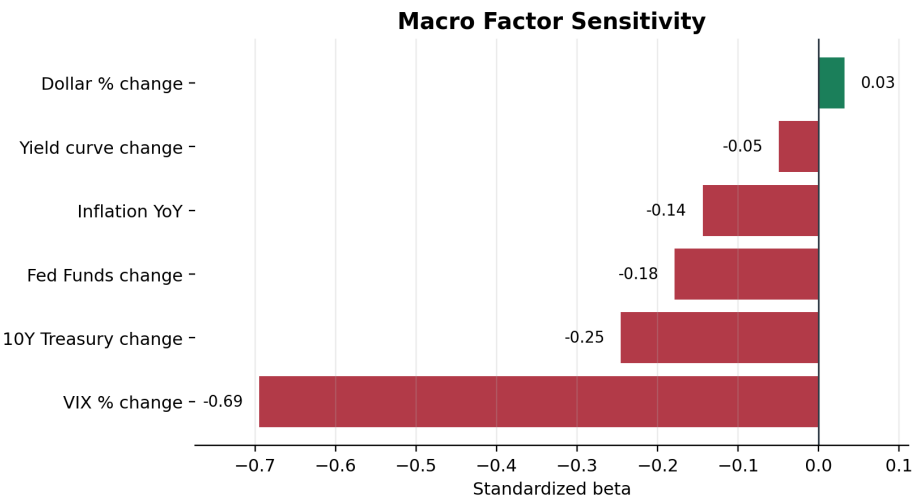
Interpretation discipline

The fan chart is useful for long-term planning because it shows dispersion and downside scenarios, not only the median outcome. However, the model assumes the historical monthly return distribution remains informative. It does not include taxes, transaction costs, regime changes, personal liquidity needs or future structural breaks.

How macro variables were created

The analysis merged monthly portfolio returns with macro variables representing rates, inflation, risk sentiment and the dollar. Raw series included Fed Funds Rate, 2Y and 10Y Treasury yields, CPI, VIX and a broad dollar index. The model then created changes and percentage changes to capture shocks rather than only levels.

Variable	Transformation	Interpretation
DGS10_Change	monthly diff.	10Y yield shock
DFF_Change	monthly diff.	Fed policy shock
Curve_Change	diff. of 10Y-2Y	yield-curve shock
Inflation_YoY	CPI pct change 12m	inflation regime
VIX_Pct_Change	VIX pct change	risk sentiment shock
Dollar_Pct_Change	dollar pct change	USD strength shock



Regression model

Monthly portfolio return was regressed on the macro factors: $R_{p,t} = \alpha + \beta_1 \text{DGS10_Change} + \beta_2 \text{DFF_Change} + \beta_3 \text{Curve_Change} + \beta_4 \text{Inflation_YoY} + \beta_5 \text{VIX_Pct_Change} + \beta_6 \text{Dollar_Pct_Change} + \text{error}$. HAC robust standard errors with 3 lags were used to reduce time-series inference bias.

Metric	Value
Observations	77 months
R-squared	0.4798
Adjusted R-squared	0.4352
Mean actual return	0.87% monthly

Why standardized betas matter

Raw coefficients use different units. Standardized betas put all variables on a comparable scale, making it easier to see that VIX shocks dominated the macro sensitivity of the portfolio.

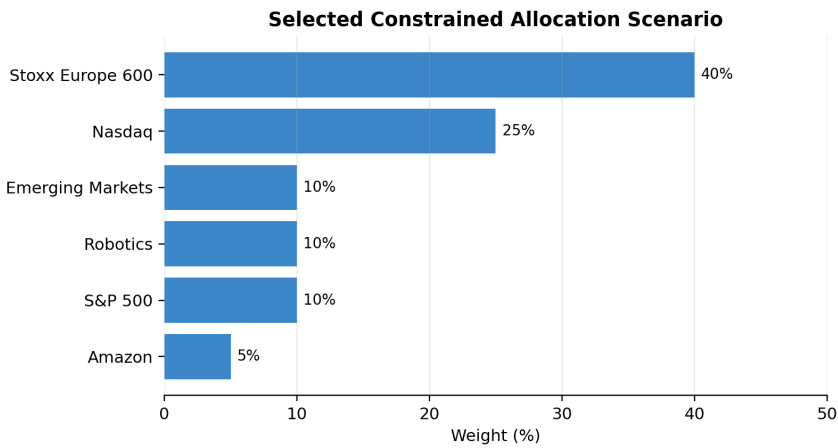
Main macro-regression results

Factor	Raw coeff.	Std beta	t-stat	p-value	Sig.
VIX % change	-0.1018	-0.6948	-8.337	0.0000	***
10Y Treasury change	-0.0452	-0.2459	-2.247	0.0246	**
Fed Funds change	-0.0324	-0.1790	-1.217	0.2238	
Inflation YoY	-0.2549	-0.1441	-0.954	0.3403	
Yield curve change	-0.0146	-0.0494	-0.517	0.6055	
Dollar % change	0.1028	0.0321	0.244	0.8070	

The result is economically intuitive for a growth-oriented equity portfolio. The strongest negative standardized beta is linked to VIX percentage changes, meaning the portfolio is highly exposed to volatility/risk-off shocks. The 10Y Treasury change is also negative and statistically significant, consistent with sensitivity to discount-rate repricing. Fed Funds, inflation, curve and dollar variables are less statistically strong in this specific sample.

How Markowitz was implemented

The allocation exercise used a grid search, not a black-box optimizer. Candidate portfolios were generated in 5% steps with minimum and maximum weight constraints. Gold was excluded from this optimization universe to test the equity/growth sleeve separately, while Amazon was fixed at 5%. For each candidate portfolio, the code computed expected monthly return, covariance-based volatility, Sharpe, Markowitz utility, max drawdown, HHI concentration and macro betas.



Score component	Weight	Direction
Markowitz utility	35%	higher is better
Markowitz Sharpe	15%	higher is better
Fed beta penalty	20%	lower abs. beta
Inflation beta penalty	15%	lower abs. beta
HHI concentration	10%	lower concentration
Max drawdown	5%	lower loss
Risk aversion lambda	3.0	variance penalty

Selected scenario metrics

Mean monthly return	Monthly volatility	Max drawdown	Beta to Fed	Beta to inflation	Final score
0.8478%	4.2799%	-22.69%	-0.0280	-0.2387	0.7776

Additional selected metrics: cumulative return 78.73%, macro R-squared 0.4918 and HHI concentration 0.2550.

What a reader should understand

- 1 The project starts from a real portfolio and turns it into a structured risk-diagnostic case study, not a simple performance report.
- 2 The portfolio had a strong equity/growth tilt. Several assets looked different by name, but behaved similarly under correlation analysis.
- 3 Individual VaR/CVaR showed that Amazon and Robotics carried the largest standalone monthly tail risk, while Gold had the lowest monthly CVaR among the assets.
- 4 The macro regression showed that VIX shocks and 10Y Treasury yield changes were the most relevant explanatory factors for portfolio returns in the sample.
- 5 The Markowitz + macro extension used a transparent scoring system that balanced return, volatility, Sharpe, drawdown, concentration and macro sensitivity.

How this project can be positioned in applications

This project is best positioned as a portfolio analytics and risk management case study. It demonstrates Python-based data handling, investment reasoning, downside-risk measurement, macro-factor modeling and constrained allocation design. It is particularly relevant for asset management, wealth management, risk management, fintech analytics and investment research roles.

Limitations

The analysis is backward looking and based on the available historical sample. The Monte Carlo simulation is distributional not predictive. The regression is only for explanatory purposes. The Markowitz extension is a

One-line CV description

Built a Python portfolio analytics framework using FactSet data to assess allocation quality, tail risk, macro sensitivity and constrained Markowitz-style allocation improvements.

PROJECT 3 / 3

Bitcoin intraday market microstructure research

Preliminary search for uncorrelated alpha using 1-minute BTC data, rare-event prediction, walk-forward validation and risk-controlled trading rules.

974k

1-minute bars
BTC-USD OHLC

50

features
microstructure proxy

30-60 min

forecast horizons
big-move labels

50 bps

event threshold
up/down targets

Core research hypothesis

The project does not try to forecast the exact Bitcoin price. It searches for rare intraday states in which the probability of a large short-term move becomes higher than the unconditional base rate. The research objective is to develop a rule-based, risk-controlled trading engine that can potentially generate alpha less dependent on passive BTC exposure.

Professional interpretation

The current evidence is promising but preliminary. Classification quality is strong relative to rare-event base rates, especially for Random Forest and MLP probability signals, but trading performance is not yet stable across all validation designs. The correct positioning is: research toward decorrelated alpha, not a production-ready strategy.

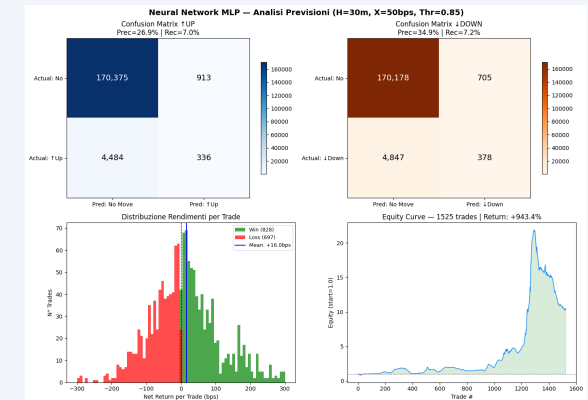


Exhibit: MLP rare-event classification and trade diagnostics

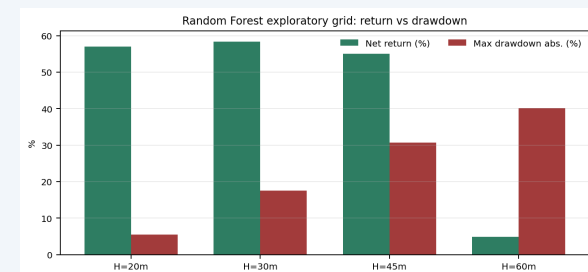
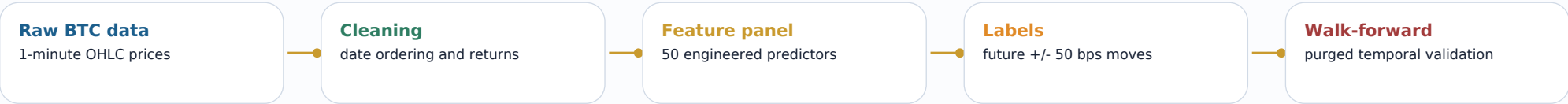


Exhibit: RF exploratory return/drawdown grid

1. Data architecture: from raw BTC bars to a no-leak feature panel



Dataset construction logic

The raw dataset contains 974,098 one-minute observations, covering the period from 2021-12-31 04:03 to 2022-12-14 00:00. The notebook creates a structured file named BTC_RF_features_noleak_1m.csv, where each row represents an investable minute and each predictor is computed using information available before the forecast horizon. This is essential because intraday models are highly exposed to look-ahead bias.

Microstructure-inspired feature set

Feature family	Examples	Market information captured
Return memory	ret_1m, logret_1...240	recent price pressure and short-term autocorrelation
Candle structure	range_pct, body_pct, wick ratios, close_pos	intrabar imbalance, rejection, pressure
Momentum	mom_mean/sum 5-120	directional acceleration across horizons
MA distance	dist_ma_15...240	trend location and distance from local fair value
Volatility regime	vol_15...240, vol_ratio	realized volatility compression/expansion
Local extremes	dist_hh/ll, squeeze_30_120	proximity to breakout/breakdown zones

Why this matters

- Minute data allows the model to focus on temporary dislocations rather than long-term BTC beta.
- Features are derived from price dynamics, volatility, candle shape and local extremes, not from fundamental data.
- The target is rare by construction, so precision lift matters more than raw accuracy.
- Embargo and walk-forward splits reduce train-test contamination.

Primary target	Big move +/- 50 bps
Date range	2021-12 to 2022-12
Feature count	50
Columns	60
Rows loaded	974,098

2. Predictive engine: rare-event classification, not price forecasting

Neural Network MLP

Deep nonlinear classifier using standardized feature input, dropout, validation split and walk-forward retraining. Best used as a probability signal; high turnover version is diagnostic, not deployable.

Random Forest

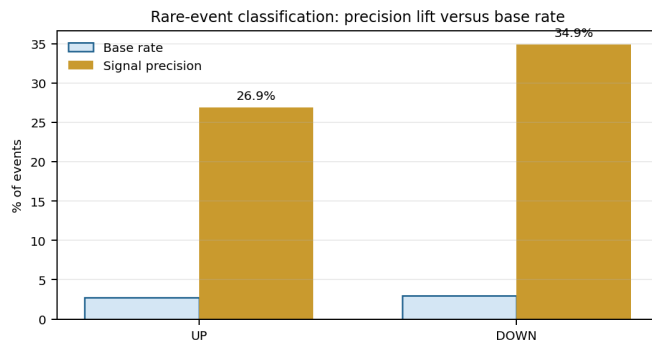
Tree ensemble classifier that captures nonlinear thresholds in volatility, momentum and local range variables. It showed the strongest exploratory trade-level results, but with configuration sensitivity.

Risk controls

The backtest includes round-trip costs, probability thresholds, cooldown, max trades/day, one-position logic and TP/SL exits. These controls convert prediction into an executable trading process.

MLP classifier diagnostic - H=30m, X=50bps, threshold=0.85

Signal	Base rate	Signals	Precision	Recall	TP	FP	Precision lift
UP $\geq$ 50 bps in 30m	2.74%	1,249	26.9%	7.0%	336	913	9.8x
DOWN $\leq$ -50 bps in 30m	2.97%	1,083	34.9%	7.2%	378	705	11.8x



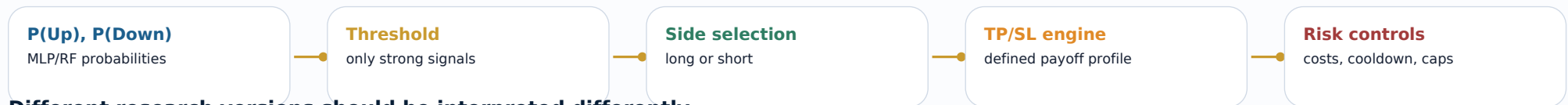
How to read the signal

The base rate of large moves is very low: only 2.74% of OOS bars trigger an UP event and 2.97% trigger a DOWN event. The MLP emits far fewer signals than bars and achieves precision of 26.9% for UP and 34.9% for DOWN. This is not high in absolute terms, but it is materially above the unconditional probability of the event. The model is therefore better interpreted as a filter that concentrates rare opportunities, not as a complete standalone trading system.

Key point

For rare-event trading, recall can be low if precision and payoff asymmetry are sufficient. The goal is to select only a small subset of high-conviction states.

3. Trading architecture: turning probability scores into controlled exposure



Different research versions should be interpreted differently

Research run	Trades	Hit rate	Mean net/trade	Net return	Professional reading
MLP H30 diagnostic	1,525	54.3%	+16.0 bps	+943.4%	High turnover / event concentration
MLP H60 professional	60	35.0%	-6.0 bps	-3.64%	Low-trade, stricter execution
RF exploratory grid	94-148	50.0%-67.0%	+31.8 to +47.9 bps	+55%-58%	Promising but short validation window

Research pipeline, not a sales pitch

The high-trade MLP run proves that the model can locate periods with strong intraday directional opportunity, but the result is concentrated and heavily exposed to regime/event risk. The stricter 60-minute professional version is more realistic, but it loses money, showing that probability quality alone is not enough. The Random Forest runs show stronger preliminary edge, but still require longer validation before being marketed as stable alpha.

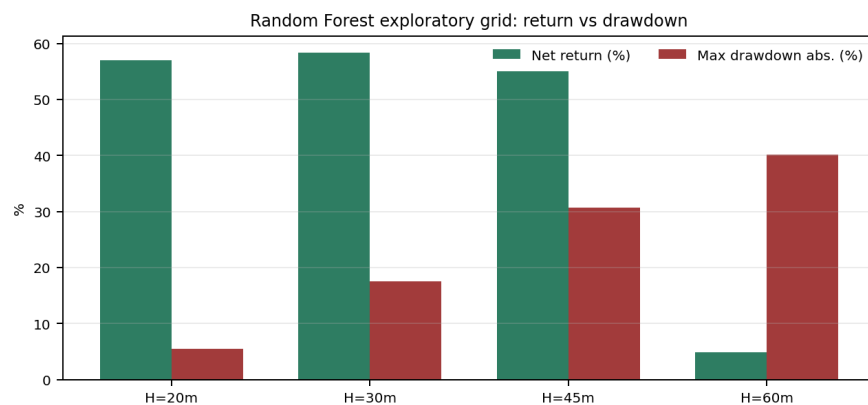
Risk-management layer

- Round-trip trading cost: 8 bps in the professional MLP run.
- TP/SL grids: 80/40, 100/50 and 120/60 bps tested.
- Cooldown: 60-120 minutes to reduce overtrading after exits.
- Max trades/day: 3-5 in the low-trade setup.
- One position at a time to avoid overlapping exposure.

4. Empirical evidence: Random Forest is promising, but robustness is not solved

Random Forest exploratory grid results

Model	H	X	Net return	Max DD	Mean/trade	Hit rate	Trades	AUC Up/Down
RF	20m	50bps	+57.07%	-5.44%	+46.5 bps	75.5%	98	0.778/0.789
RF	30m	50bps	+58.38%	-17.48%	+31.8 bps	50.0%	148	0.748/0.749
RF	45m	50bps	+55.09%	-30.67%	+47.9 bps	67.0%	94	0.724/0.720
RF	60m	50bps	+4.86%	-40.22%	+6.9 bps	60.5%	86	0.694/0.698



Interpretation of results

The RF grid shows that the model can identify high-probability microstructure states with positive average trade return. However, results are sensitive to horizon and holding time. H=20m and H=30m deliver similar net return with materially different drawdown; H=45m keeps high mean return but increases drawdown; H=60m loses most of the edge. This suggests that the signal is short-lived and should be treated as an intraday microstructure effect rather than a broad directional BTC forecast.

Annualized metrics are intentionally not emphasized: the test window is short and intraday strategies can produce misleading annualized ratios.

5. Microstructure reading: the edge appears event-driven and regime-dependent

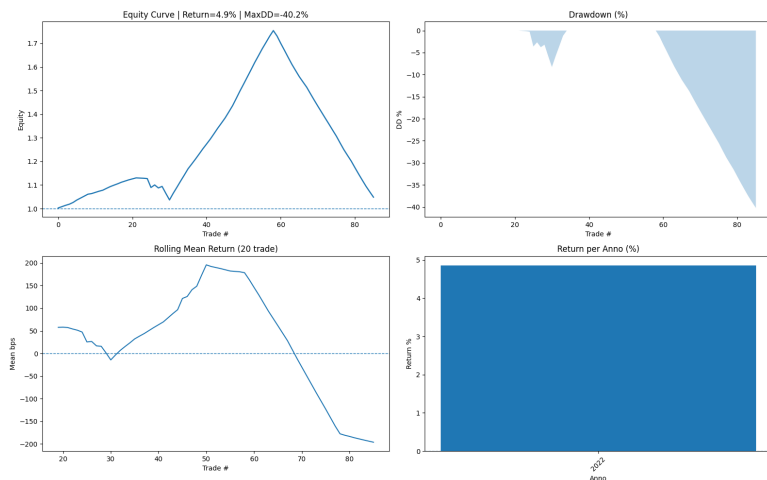


Exhibit: robustness check from trade log - equity, drawdown, rolling mean and annual return

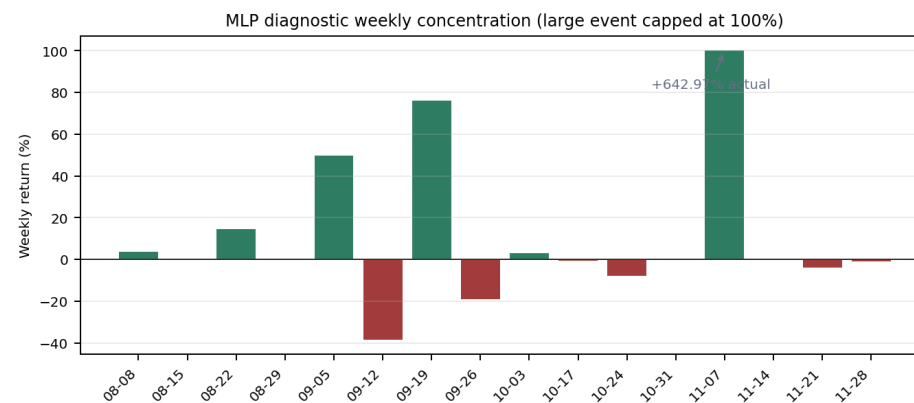


Exhibit: diagnostic weekly concentration - regime dependence and event risk

What the strategy is actually detecting

The feature set is microstructure-inspired: short returns, candle shape, wick pressure, range compression, volatility ratios, moving-average distance and distance from recent highs/lows. These variables attempt to detect when the order-flow environment is no longer neutral, even without direct order-book data.

The most important limitation is stability. The model can identify profitable clusters, but those clusters are not evenly distributed. This is consistent with intraday crypto markets, where liquidity, news shocks and forced deleveraging can temporarily dominate normal price formation.

Market pressure

Volatility expansion

Local breakout states

Regime dependence

6. Final assessment: credible research toward decorrelated alpha

The notebook demonstrates a real quantitative research process: data engineering, no-leak feature construction, supervised learning, walk-forward testing, trading rules and risk diagnostics.

What works

Rare-event classifiers show meaningful lift versus the base probability of large intraday moves. Random Forest reaches AUC values around 0.72-0.79 across configurations, and selected RF runs generate positive net exploratory results.

What is not solved

The alpha is not yet stable enough to be presented as deployable. Some performance is concentrated in specific weeks, model configurations produce different outcomes, and the professional low-trade MLP version is negative after costs.

Why it is valuable

The project demonstrates the ability to build a full intraday ML trading research pipeline and to interpret results critically rather than overfitting a backtest. This is highly relevant for quant research and risk-managed portfolio construction.

Research roadmap to make the alpha claim stronger

Next development	Purpose	Professional rationale
Longer sample and multi-exchange data	Test 2023-2026 and exchange robustness	reduce sample and venue-specific bias
BTC benchmark and correlation analysis	Measure beta to BTC buy-and-hold	prove decorrelation instead of assuming it
Real execution model	Slippage, bid-ask, latency, fees	convert paper edge into tradable edge
Regime filter	Trade only in suitable volatility/liquidity states	reduce drawdown and unstable weeks
Position sizing	Volatility targeting and drawdown limits	stabilize P&L path and capital usage
Feature upgrade	Order book, volume, funding, liquidations	move from proxy microstructure to actual microstructure

Portfolio takeaway

This is best presented as a preliminary, risk-aware intraday research framework for finding market-independent alpha - not as final strategy. Its value lies in the full pipeline and the honest robustness analysis.